

ODENSE AP, Denmark

WMO: 061200

Lat: 55.475N

Lon: 10.331E

Elev: 56

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.6	22.4	14.1	11.3	22.6	17.7	13.4	24.7	33.8	43.7	30.7	44.5	6.9	120	0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	80.7	66.2	77.2	64.9	73.9	63.5	69.0	75.6	67.0	73.4	65.3	71.4	10.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.7	98.3	71.5	64.7	91.7	69.6	63.0	86.2	67.7	33.1	75.6	31.5	73.3	30.2	71.5	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.1	23.9	21.4	DB	10.9	85.7	8.1	3.8	5.1	88.4	0.4	90.6	-4.2	92.8	-10.1	95.5	(4)
(5)			WB	10.6	72.6	7.8	2.3	5.0	74.3	0.4	75.6	-4.0	76.9	-9.6	78.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	34.7	35.5	38.9	45.6	52.9	58.5	63.1	63.0	57.1	49.4	42.3	37.0
	DBStd	11.64	6.64	6.76	5.85	5.70	5.34	4.92	4.73	4.55	4.38	5.29	5.66	7.13
	HDD50	2127	474	406	344	154	31	1	0	0	3	74	238	402
	HDD65	6197	939	826	808	581	376	203	97	97	238	483	682	867
	CDD50	1485	0	0	1	23	121	256	405	401	216	56	5	1
	CDD65	81	0	0	0	0	1	8	37	33	2	0	0	0
	CDH74	712	0	0	0	4	16	98	334	249	11	0	0	0
	CDH80	117	0	0	0	0	1	15	57	44	0	0	0	0
Wind	WSAvg	10.8	12.2	12.6	12.2	10.8	10.0	10.0	9.3	9.5	10.2	10.6	11.0	11.5
Precipitation	PrecAvg	25.7	2.4	1.7	1.5	1.3	1.8	2.5	2.3	2.8	2.4	2.6	2.2	2.3
	PrecMax	32.2	4.6	3.8	3.5	3.3	3.1	5.7	4.6	5.5	5.6	6.4	5.1	4.9
	PrecMin	17.8	0.1	0.2	0.3	0.1	0.7	0.1	0.4	0.9	0.7	0.9	0.5	0.7
	PrecStd	3.9	1.2	0.8	0.8	0.7	0.7	1.3	1.2	1.4	1.3	1.1	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.7	51.5	57.6	70.2	75.9	82.2	85.0	84.2	75.5	65.8	57.5	52.2
		MCWB	48.2	47.8	48.8	57.8	63.3	68.0	67.7	67.0	63.8	59.5	56.2	50.8
	2%	DB	47.2	48.3	53.4	63.9	70.5	76.6	81.0	80.5	71.5	61.1	54.3	50.1
		MCWB	46.0	46.4	47.2	53.5	59.2	64.9	66.0	65.8	62.2	57.6	52.9	48.9
	5%	DB	45.3	46.3	49.8	59.3	67.0	72.3	77.7	76.3	67.8	59.1	52.1	48.0
		MCWB	44.2	44.8	45.8	51.2	57.7	62.5	65.1	64.7	60.8	56.5	50.7	47.0
	10%	DB	43.5	44.4	47.3	55.7	63.8	69.1	74.4	72.9	64.8	57.2	50.1	46.2
		MCWB	42.5	42.9	44.5	49.5	55.8	60.6	63.8	63.6	59.3	55.1	48.9	45.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.5	48.9	51.4	59.4	66.1	71.9	71.8	70.8	66.7	61.1	56.7	51.2
		MCDB	49.1	50.0	54.9	67.7	73.0	80.5	78.1	77.3	72.0	64.2	57.2	51.7
	2%	WB	46.2	46.6	48.4	55.2	62.0	67.1	69.1	68.5	63.9	58.7	53.2	49.2
		MCDB	46.9	47.7	51.4	61.1	67.2	73.2	76.7	73.9	68.0	60.7	54.1	50.0
	5%	WB	44.3	44.8	46.5	52.6	59.4	64.0	67.2	66.6	62.0	56.9	51.1	47.1
		MCDB	45.0	45.9	48.7	57.7	64.5	69.8	74.2	72.7	65.9	58.6	52.0	47.9
	10%	WB	42.5	42.9	44.9	50.4	57.0	61.8	65.3	65.0	60.5	55.4	49.0	45.2
		MCDB	43.3	44.1	46.9	54.5	61.9	67.2	71.7	70.9	64.1	57.0	49.9	46.0
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.9	11.2	14.8	16.2	16.4	17.1	16.1	13.2	10.6	7.9	7.3
		MCDBR	7.7	9.1	15.4	21.0	22.4	22.3	23.1	22.1	17.7	12.3	8.4	7.4
		MCWBR	7.4	7.3	10.4	11.9	13.1	12.7	11.0	10.3	10.3	9.3	7.4	7.2
	5% WB	MCDBR	7.6	8.5	13.2	18.7	19.3	19.9	20.1	18.5	15.6	11.5	7.8	7.1
		MCWBR	7.4	7.3	9.8	11.8	12.9	12.9	11.1	10.6	10.1	9.2	7.3	7.1
Clear-Sky Solar Irradiance	taub	0.318	0.338	0.363	0.398	0.391	0.384	0.403	0.403	0.378	0.357	0.335	0.305	
	taud	2.274	2.303	2.285	2.238	2.329	2.375	2.339	2.351	2.402	2.406	2.327	2.252	
	Ebn at Noon	187	230	253	260	270	273	265	257	247	222	180	162	
	Edh at Noon	18	25	32	38	37	36	36	34	28	23	17	15	
All-Sky Solar Radiation	RadAvg	147	339	790	1313	1686	1807	1678	1342	950	477	187	105	
	RadStd	25	52	74	156	161	132	179	91	93	60	29	10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (63 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	-209	-304	N/A	N/A

Nomenclature: See separate page